

JIHANG LI

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SUMMARY

- Software engineer at LaunchPath, where I shipped an AI sales-outreach platform end to end — Spring Boot, React/TypeScript and PostgreSQL, with RAG-grounded generation on pgvector and autonomous LLM agent workflows.
- Graduate researcher at UC Irvine, designing C++ Monte Carlo and PCA algorithms for automated 3D cell-lineage reconstruction, with a research manuscript in progress.
- Machine learning and computer vision in PyTorch and TensorFlow — Gaussian-splatting scene reconstruction and object detection for autonomous driving in the AIEA Lab at UC Santa Cruz, and PPO reinforcement-learning agents trained in GPU Docker.

EDUCATION

UNIVERSITY OF CALIFORNIA, IRVINE, Irvine, California
Master of Computer Science, Expected Dec 2026 · GPA: 3.97/4.00

UNIVERSITY OF CALIFORNIA, SANTA CRUZ, Santa Cruz, California
B.S., Computer Science, Sep 2021 – Dec 2024 · GPA: 3.67/4.00

TECHNICAL SKILLS

Languages:	Python, C/C++, Java, TypeScript/JavaScript, SQL, HTML/CSS
Frameworks:	Spring Boot, React, Node.js, Django, REST API
Developer Tools:	Git, Docker, Google Cloud Platform, VS Code, Visual Studio, PyCharm, IntelliJ, Jupyter Notebook
Databases & Infra:	PostgreSQL, pgvector, Flyway, Testcontainers
Libraries:	PyTorch, TensorFlow, CUDA, Stable-Baselines3, Pandas, NumPy, Matplotlib

EXPERIENCE

LaunchPath — Software Engineer Intern, LaunchMail · Remote, California **Jun 2026 – Present**

- Shipped LaunchMail, an AI sales-outreach platform on Spring Boot, React/TypeScript and PostgreSQL, with 110+ REST endpoints.
- Implemented RAG-grounded AI email drafting with pgvector embeddings, unified across 5 LLM providers.
- Built autonomous outreach agents and a drag-and-drop workflow engine on a webhook-resumed Postgres state machine.
- Engineered deliverability safeguards: sender rotation, throttling, daily caps and a 220-term spam linter, shielding customer domains.
- Ensured reliability with 242 integration tests (Testcontainers, WireMock) across 33 Flyway schema migrations.

UC Santa Cruz — AIEA Lab — Undergraduate Researcher, under Dr. Leilani Gilpin **Jan 2024 – Jun 2025**

- Researched Gaussian splatting techniques to reconstruct dangerous driving scenes for vehicle hazard-response analysis.
- Conducted object detection research to improve recognition in complex urban areas using machine learning algorithms.
- Utilized Python, TensorFlow and PyTorch for experiments, data analysis, model development and evaluation.
- Collaborated with a multidisciplinary team to advance state-of-the-art research in autonomous driving.
- Improved Gaussian-splatting rendering efficiency, accelerating playback of reconstructed hazard scenes for analysis.

PROJECTS

CellUniverse — 3D Cell Tracking · C++ · Algorithms · MCS Keystone research under Dr. Wayne Hayes **Feb 2026 – Present**

- Replaced voxel-matrix rendering with analytical inverse-rotation ellipsoid testing, cutting memory and fixing artifacts.
- Built a Monte Carlo optimizer fitting 3D oblate spheroids to confocal microscopy data through iterative L2 cost minimization.
- Developed PCA-based cell division detection, clustering real-image bright pixels to locate split axes and daughter-cell positions.
- Implemented per-axis normalization and neighbor-exclusion filtering to prevent PCA contamination from adjacent cells.
- Designed two-phase optimization separating cell fitting from split detection, eliminating cascading false divisions.
- Contributing to a research manuscript on automated 3D cell lineage tracking across imaging datasets.

Doom RL Agent · Reinforcement Learning · Python · PPO, Stable-Baselines3, VizDoom **Sep 2025 – Dec 2025**

- Implemented PPO agents via Stable-Baselines3 targeting two VizDoom scenarios with increasing enemy complexity.
- Designed a reward shaping system with survival bonuses, death penalties and kill rewards to guide exploration.
- Applied running-statistics reward normalization to stabilize gradients and accelerate policy convergence across scenarios.
- Tuned PPO hyperparameters (learning-rate decay, batch size, entropy coefficient), lifting survival from ~38 to 300 steps.
- Containerized training with a GPU-accelerated Docker environment using CUDA 11.8 and docker-compose for reproducible runs.